

The S-Curve After the Shock: Machine Learning Evidence on Mortgage Prepayment in the Lock-In Era

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Abstract

We estimate loan-level mortgage prepayment hazard models (logistic regression, gradient-boosted trees, and a deep neural network in the spirit of [Sadhvani et al. \(2021\)](#)) on 18.5 million Fannie Mae loans observed monthly between 2018 and 2025, and use them to characterize how the prepayment function changed after the 2022 interest-rate shock. A model estimated through 2021 over-predicted aggregate speeds by roughly 9 CPR at the onset of the shock, and its errors never recovered: between 3.1 and 3.8 CPR points of cohort-level error in every year through 2025. Annual re-estimation halved the error within a year, then needed about two more years of data to get back below one point. The prepayment function itself did not flatten in the lock-in era, as is sometimes argued; it rotated. Holding borrower composition fixed, the refinancing response above the money is more than twice as steep as before the pandemic (30.5 versus 13.2 CPR at +150 basis points of incentive), and the slow speeds of discount coupons are carried by the level of prevailing rates rather than by a weaker incentive response. Applied to the December 2025 cross-section, the model places 2023–24 vintage 7.0% coupons near 25 CPR at unchanged rates and above 50 CPR in a 150 basis point rally. A caution for empirical work on this dataset: several fields are blanked on removal records and can leak the prepayment event into standard covariates, which, left untreated, inflates out-of-sample AUC to 0.9999. All code and artifacts are public.

Keywords: mortgage prepayment; MBS; machine learning; lock-in effect; gradient boosting; interpretability.

1 Introduction

Between March 2022 and October 2023 the 30-year fixed mortgage rate in the United States rose from below 4% to nearly 8%, the fastest repricing of the mortgage market in four decades. Aggregate conditional prepayment rates (CPR) on Fannie Mae collateral fell from above 35 to

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below 5, and by the end of 2023 nearly the entire agency universe was out of the money. The episode placed prepayment models, the central pricing input for a \$9 trillion market, outside the support of their training data. The practitioner record of the aftermath is extensive: vendor models missed monthly aggregate prints by 10–25% through 2023–2025, disagreed on the sign of convexity below par, and generated relative-value signals that inverted against realized returns (Yu, 2023; Katz and Li, 2024).

A parallel academic literature has quantified the underlying mechanism, rate lock-in, from the housing side. Batzer et al. (2024) estimate that each percentage point by which prevailing rates exceed a borrower’s contract rate reduces the probability of sale by 18.1%; Fonseca and Liu (2024) find a 9–16% reduction in moving probability per percentage point of lock-in; Liebersohn and Rothstein (2025) attribute a 16% decline in the mobility of mortgaged households to the mechanism. This literature measures the turnover component of prepayment, but does not address the model-side question that concerns fixed-income practitioners: *what exactly changed in the prepayment function, and how should data-driven models be managed across such a break?*

This paper takes up both questions on open data. We construct a monthly discrete-time hazard panel from the Fannie Mae Single-Family Loan Performance dataset (18.5 million loans, 2018–2025), estimate three classes of prepayment model of increasing flexibility under a strict walk-forward protocol, and evaluate them at two levels: loan-level discrimination and, more importantly for economic purposes, UPB-weighted cohort-level CPR error against full-population actuals.

The first set of results concerns model governance across the break (Section 5.2). The pre-shock model fails immediately, and its errors never mean-revert: 3.1 to 3.8 CPR points of UPB-weighted cohort error in every year from 2022 through 2025. A structurally broken model, in other words, could not be “ridden out.” Re-estimation worked, but slowly: annual refits improve steadily and still need roughly two years of new-regime data before cohort errors fall below one CPR point, a pace of recovery that matches what has been documented for commercial models over 2023–25.

The second set of results concerns the shape of the prepayment function (Section 5.3). Estimating the model separately by regime and tracing each fit’s partial-dependence S-curve on a fixed reference population separates composition from behavior, and the answer is not the flattening that is sometimes claimed. Conditional on composition, the refinancing limb of the lock-in-era S-curve is *steeper* than its pre-pandemic counterpart: the elasticity through the cusp rises from 2.2 to 12.6 CPR points per 100 basis points. Depressed deep-discount turnover, meanwhile, is absorbed by the prevailing-rate level rather than by a decayed incentive response. Both sides of a live practitioner debate (Landy, 2024; Katz and Li, 2024) turn out to be describing the same estimated function.

Finally, the estimates translate into a forward-looking statement about the current coupon stack (Section 5.4). As of December 2025, model-implied speeds for the 2023–24 vintage 6.5–7.0% coupons rise steeply under modest rallies; the 6.5s roughly triple between unchanged rates and a 150 basis point rally.

A secondary, methodological contribution concerns data quality. Three fields in the public dataset are blanked on a loan’s terminal record, so that, if used contemporaneously, their miss-

ingness perfectly reveals the prepayment event. Untreated, these produce out-of-sample AUCs of 0.9999 and would invalidate any study built on them. Section 3.4 documents the mechanism and the corrections, which we regression-test in the released code.

2 Related literature

Machine learning for mortgage risk. [Sadhvani et al. \(2021\)](#) estimate multi-state transition models on 120 million U.S. mortgages with deep neural networks and document large out-of-sample gains over logistic benchmarks, with prepayment exhibiting the strongest nonlinearities. [Zhang et al. \(2019\)](#) deploy neural networks for agency MBS prepayment at pool level and interpret the fitted model through implied S-curves, burnout, and seasonality; [Schultz and Fabozzi \(2021\)](#) make the case for gradient-boosted trees at loan level. [Blumenstock et al. \(2022\)](#) find machine-learning survival models dominate statistical benchmarks for competing default and prepayment risks; [Ojha and Lee \(2021\)](#) compare conventional and deep learners on Fannie Mae data through the financial crisis. Relative to this literature, our contribution is not architectural novelty but the regime question: all published samples of which we are aware end before or shortly after the 2022 break, and none addresses model management across it.

The lock-in effect. [Batzer et al. \(2024\)](#), [Fonseca and Liu \(2024\)](#), and [Liebersohn and Rothstein \(2025\)](#) establish the magnitude of rate lock-in for housing turnover, labor mobility, and welfare. We use their elasticities as external anchors for the turnover-suppression channel recovered by our models.

Practitioner record. [Yu \(2023\)](#) frames post-2022 prepayment modeling as an out-of-sample problem; monthly commentary by Santander US Capital Markets documents production model errors of 10–25% through 2023–25 and argues that apparent S-curve flattening reflects spread-at-origination (SATO) composition ([Landy, 2024](#)); [Katz and Li \(2024\)](#) document model-implied relative value inverting against realized returns in 2024 and, in early 2026, pool-level S-curves exceeding pandemic-era steepness. Our per-regime, composition-controlled S-curves reconcile these observations within a single estimated model.

3 Data

3.1 Loan-level performance data

We use the Fannie Mae Single-Family Loan Performance dataset, acquisition quarters 2018Q1 through 2025Q4: 18,468,917 fixed-rate loans and their complete monthly servicing records through December 2025, parsed from the raw 113-field pipe-delimited files. The window brackets three regimes: pre-pandemic (2018–19), the pandemic refinancing wave (2020–21), and the lock-in era (2022–25). Voluntary prepayment is identified by zero-balance code 01 (payoff); credit events and repurchases are treated as censoring.

3.2 Cohort-level ground truth

From the full population, prior to any sampling, we compute actual cohort-level single monthly mortality and CPR by vintage year and coupon (rounded to the nearest 0.5), where SMM is voluntarily prepaid UPB over beginning-of-month UPB. These series serve as the evaluation target for all models and align with public anchors: 2023-vintage 7.0 coupons print 5.4 CPR in June 2024, peak at 30.3 in the October 2024 refinancing episode, and revert to 9.1 by December ([Federal Housing Finance Agency, 2024](#)); 2021-vintage 2.5 coupons remain at 2–4 CPR throughout 2023–25; 2018-vintage 4.5 coupons run at 41–57 CPR through the pandemic wave ([Haughwout et al., 2023](#)).

3.3 Estimation panel

A deterministic, hash-based stratified sample (vintage year $\times$ note-rate bucket, with inverse-probability loan weights) of 1,201,243 loans yields a monthly hazard panel of 2,532,987 loan-month observations containing all 428,793 voluntary prepayment events and a 5% importance-weighted subsample of non-event months. Weights are carried through estimation and aggregation, so fitted hazards are on the monthly SMM scale and cohort aggregates are population-representative.

Covariates comprise: the refinancing incentive (note rate minus the Freddie Mac PMMS 30-year survey rate); spread at origination (SATO); burnout (cumulative positive incentive over prior months); mark-to-market LTV (original LTV revalued with state-level FHFA house-price indices); loan age; original balance; credit score; DTI; original LTV; calendar month; the prevailing mortgage-rate level; and categorical origination attributes (channel, purpose, property type, occupancy, state, borrower count, first-time-buyer flag) plus lagged delinquency status and modification flag.

3.4 Removal-record measurement pitfalls

Three covariates initially produced out-of-sample AUCs of 0.9999, a value that should be read as diagnostic of leakage rather than of predictive skill. The mechanism is systematic: the dataset blanks monthly-reported fields on a loan’s terminal record. Consequently (i) mark-to-market LTV computed on the end-of-period balance equals exactly zero if and only if the loan prepaid that month; (ii) the modification flag and delinquency status are missing on removal records, so their missingness indicators reveal the event; and (iii) the reported loan-age field is likewise blanked. All state-dependent inputs are therefore measured as of the *beginning* of the month: balances lagged (the factor-date convention), status fields lagged one month, and loan age computed from calendar arithmetic rather than read from the file. Each correction carries a regression test in the released code. Post-correction discrimination falls to the 0.59–0.75 range, which we regard as the honest difficulty of monthly voluntary-prepayment classification.

4 Methodology

4.1 Hazard framework and inverse-probability weighting

Let $y_{it} \in \{0, 1\}$ indicate voluntary prepayment of loan i in month t . We model the discrete-time hazard $h_{it} = \Pr(y_{it} = 1 \mid x_{it})$, where x_{it} contains only information available at the beginning of month t .

The estimation panel is drawn from the population in two independent, deterministic stages, and every observation carries the inverse of its inclusion probability. Stage one samples *loans*: within stratum $s(i)$ (vintage year $\times$ note-rate bucket), loan i is included with probability $\pi_{s(i)} = \min\{1, \bar{n}/N_{s(i)}\}$, where N_s is the stratum size and $\bar{n}$ the per-stratum target, implemented by a hash of the loan identifier (no random-number generator, hence exactly replicable). Stage two samples *loan-months*: every event month ($y_{it} = 1$) of a sampled loan is retained with probability one, while non-event months are retained with probability $\kappa = 0.05$, again by hashing. The combined design weight is therefore

$$w_{it} = \begin{cases} 1/\pi_{s(i)}, & y_{it} = 1, \\ 1/(\pi_{s(i)} \kappa), & y_{it} = 0. \end{cases} \quad (1)$$

For any population functional expressible as a sum over loan-months, $S = \sum_{it} z_{it}$, the weighted sample sum $\hat{S} = \sum_{it \in \mathcal{S}} w_{it} z_{it}$ is a Horvitz–Thompson estimator and is unbiased, since each term’s inclusion probability is exactly $\pi_{s(i)}$ or $\pi_{s(i)} \kappa$ by construction. Two consequences follow. First, maximizing the w -weighted log-likelihood targets the population likelihood, so fitted hazards are calibrated to the population base rate: retaining all events while down-sampling non-events does *not* require the usual post-hoc intercept correction for case-control sampling, because the weights already undo the design. Second, population cohort-level speeds are recovered by weighting fitted hazards with the product of design weight and beginning-of-month balance b_{it} : for cohort c in month t ,

$$\widehat{\text{SMM}}_{ct} = \frac{\sum_{i \in c} w_{it} b_{it} \hat{h}_{it}}{\sum_{i \in c} w_{it} b_{it}}, \quad \widehat{\text{CPR}}_{ct} = 1 - (1 - \widehat{\text{SMM}}_{ct})^{12}, \quad (2)$$

where numerator and denominator are each Horvitz–Thompson estimators of their population counterparts (expected voluntarily-prepaid balance and total beginning balance). Evaluation never relies on the sampled denominator where the population value is available: actual cohort CPRs are computed on the *full* population of Section 3.2. Appendix A verifies empirically that results are insensitive to halving κ , and that fitted hazards are well calibrated against weighted empirical event rates.

4.2 Estimators

We estimate three models of increasing flexibility on identical features and weights: (i) weighted logistic regression (median-imputed numerics, one-hot categoricals), the linear benchmark; (ii) gradient-boosted trees (LightGBM; 63 leaves, 600 trees, learning rate 0.05, minimum 100 observations per leaf), with missing values handled natively; and (iii) a feed-forward neural network with categorical embeddings (three hidden layers, 128–128–64 ReLU units, dropout 0.1), trained

with weighted cross-entropy, following the architecture spirit of [Sadhvani et al. \(2021\)](#). We deliberately avoid early stopping on trailing-window validation sets: in the presence of regime shifts, a trailing window is unrepresentative of both past and future, and we found it induces severe underfitting precisely at the breaks.

4.3 Evaluation protocol

All evaluation is strictly walk-forward: models are estimated on months up to December of year T and evaluated on the following twelve months, for $T = 2018, \dots, 2024$. We report loan-level weighted AUC, and UPB-weighted mean absolute cohort CPR error (in CPR points) against the full-population actuals of Section 3.2. We emphasize the latter: loan-level discrimination metrics have limited economic meaning for portfolio speeds; as Section 3.4 illustrates, they are also the metrics most easily corrupted by leakage.

5 Results

5.1 Walk-forward performance

Table 1 reports the walk-forward record. The gradient-boosted model beats the logistic benchmark on cohort-level error in six of seven windows, and the gap is widest exactly where it matters for the questions of this paper: 0.9 versus 6.1 CPR points at the 2024 split. Neither model gets the *level* of the 2020–21 refinancing wave right. Commercial models did not either; the determinants of wave amplitude (origination capacity, appraisal waivers, media effects) are not spanned by loan-level covariates, and we treat this as a limitation rather than attempting to tune it away. The AUC column also deserves comment: discrimination deteriorates for every model in the lock-in era. With the entire universe out of the money, who prepays in a given month is close to idiosyncratic, so a low lock-in-era AUC is a property of the environment, not a defect of any particular learner. As for the neural network, evaluated at three splits it is competitive on discrimination (AUC 0.75/0.65/0.64) and ahead of the trees on cohort error at the pre-wave split (8.8 versus 12.2 CPR points), but it falls behind in the lock-in era (4.6 versus 3.7, and 1.5 versus 0.9) and under-predicts the hazard level. At 2.5 million training rows a mixed result is what the tabular-learning literature would predict; the deep-learning advantage in [Sadhvani et al. \(2021\)](#) was demonstrated on three orders of magnitude more data. The gradient-boosted model is therefore the engine for the experiments that follow.

5.2 The regime break: error persistence and the pace of recovery

We compare a model frozen at December 2021 against annual re-estimation, both walked forward over 2022–25 (Figure 1). The onset is severe: in January 2022 the frozen model predicts aggregate speeds of approximately 19 CPR against an actual print of 11, an error of nearly 9 CPR points, because it extrapolates refinancing-wave behavior into a market that has just repriced. Over 2022 as a whole its UPB-weighted cohort error is 3.7 CPR points. The annual errors show that the frozen model does not improve thereafter: 3.8 points in 2023, 3.3 in 2024, 3.1 in 2025. The errors do not decay as the lock-in era matures, because the model’s learned response

Training window ends	Loan-level AUC		Cohort CPR MAE (pts)	
	Logistic	GBM	Logistic	GBM
2018-12	0.66	0.71	11.6	6.4
2019-12	0.75	0.78	12.7	12.2
2020-12	0.72	0.75	8.6	13.1
2021-12	0.61	0.66	4.6	3.7
2022-12	0.52	0.63	2.6	2.2
2023-12	0.57	0.66	5.2	0.9
2024-12	0.59	0.69	6.1	0.9

Table 1: Walk-forward evaluation. Each row: estimation through December of the stated year, evaluation on the subsequent twelve months. Cohort CPR MAE is UPB-weighted across vintage $\times$ coupon cohorts against full-population actuals. The 2020–12 test window is the peak of the pandemic refinancing wave.

surface—not merely its level—is wrong for the new regime; Figure 1 shows it running 2–4 points above realized speeds in essentially every month after mid-2022. Re-estimation recovers, but only as new-regime observations accumulate: folding in the 2022 transition year nearly halves the 2023 error (2.2 versus 3.8 points), and with two full years of post-break observations the annually re-estimated model reaches 0.9 points in both 2024 and 2025, comparable to pre-break accuracy. The governance implication is twofold: a structurally broken model cannot be ridden out in the hope that its errors mean-revert, and recalibration, while effective, is gradual. Both implications are consistent with the documented difficulties of commercial models over 2023–25 (Yu, 2023; Katz and Li, 2024).

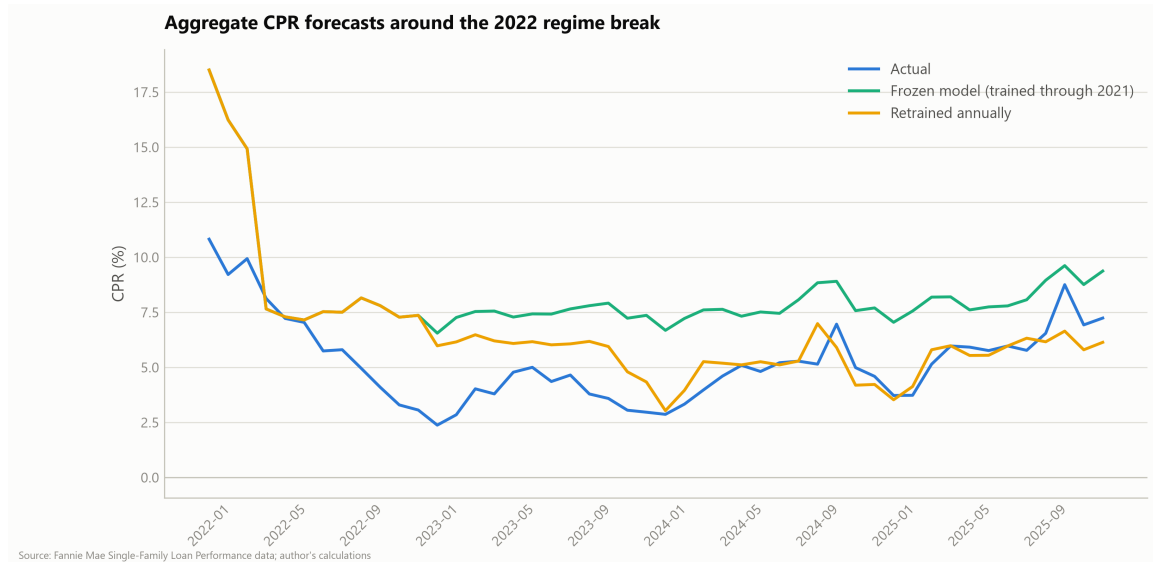


Figure 1: Aggregate UPB-weighted CPR, 2022–2025: actuals versus a model frozen at December 2021 and a model re-estimated each year-end. The two model series coincide during 2022 by construction. The frozen model over-predicts by ≈ 9 CPR points at the January 2022 onset and remains 2–4 points above realized speeds through 2025; the re-estimated model converges to realized speeds from 2024 onward.

5.3 The learned S-curve, by regime

We estimate the gradient-boosted model separately on each regime (pre-pandemic, 2018-01 to 2019-12; refinancing wave, 2020-04 to 2021-12; lock-in, 2022-07 to 2025-12) and trace each fitted model’s partial-dependence function of the refinancing incentive on a *common* reference population of 200,000 lock-in-era loan-months. Holding the population fixed ensures that differences between curves reflect learned behavior rather than composition. Figure 2 and Table 2 report the results.

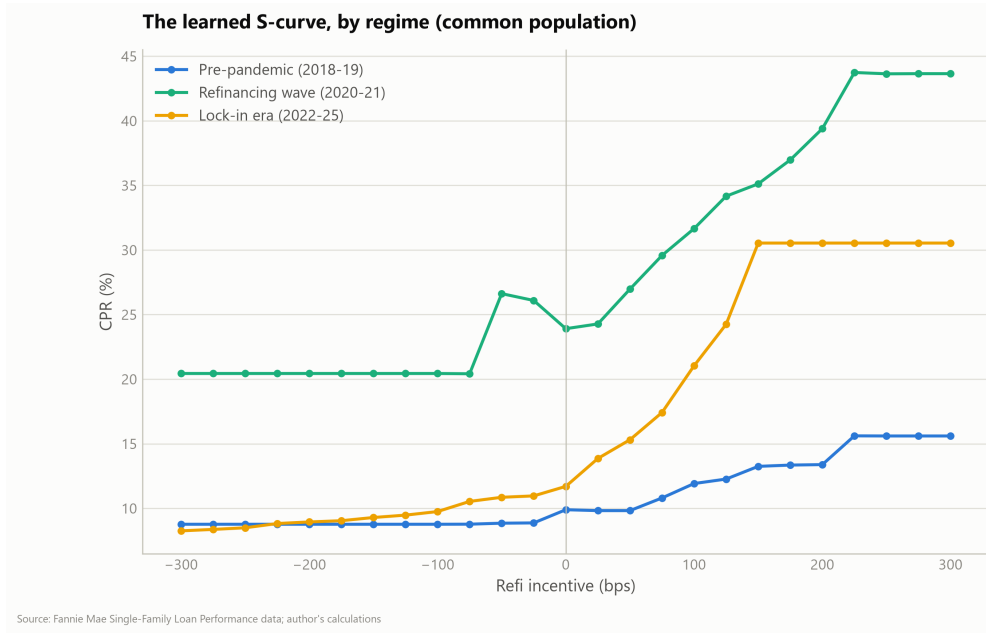


Figure 2: Model-implied S-curves (partial dependence of annualized speed on refinancing incentive) by estimation regime, evaluated on a common lock-in-era reference population.

Regime	CPR at −200bp	CPR at 0	CPR at +150bp	Elasticity (pts/100bp)
Pre-pandemic 2018–19	8.8	9.9	13.2	2.2
Refinancing wave 2020–21	20.4	23.9	35.1	7.5
Lock-in 2022–25	8.9	11.7	30.5	12.6

Table 2: S-curve landmarks by regime on the common reference population. Elasticity is the average slope between 0 and +150 basis points of incentive. All entries are partial-dependence values: counterfactual model responses obtained by setting the incentive to the stated level for every observation in the fixed reference population. They are not realized cohort speeds and should not be compared directly to observed cohort CPRs (see text).

The central finding is a rotation, not a flattening. The refinancing limb of the lock-in-era S-curve is markedly steeper than its pre-pandemic counterpart: 30.5 versus 13.2 CPR at +150 basis points, an elasticity through the cusp of 12.6 versus 2.2 CPR points per 100 basis points. Borrowers who reach the money in the lock-in era refinance faster than any pre-2020 cohort would have, consistent with the 15-month average age at refinancing observed in the 2024 episode and with the efficiency of modern origination channels. At the deep-discount end, the lock-in model’s implied turnover on the common population is close to pre-pandemic levels; the observed 2–4 CPR prints of deeply out-of-the-money cohorts are reproduced through the

prevailing-rate-level covariate, the lock-in channel of [Batzer et al. \(2024\)](#), rather than through a degraded incentive response.

Interpreting Table 2 requires care. Partial-dependence values at a given incentive are averages over the *entire* reference population with only the incentive altered; every other covariate, including burnout, mark-to-market LTV, and the prevailing-rate level, keeps its observed value. The population composition at, say, -200 basis points in the table never occurs in any realized cohort. In particular, the tabulated 8.9 CPR for the lock-in model at -200 basis points is not comparable to the 2–4 CPR prints of 2021-vintage 2.5% coupons: those cohorts sit at -300 to -450 basis points of incentive, at or beyond the left edge of the evaluated grid where the curves are lower, and their suppressed turnover loads on the rate-level covariate rather than on the incentive itself. And because the reference population is held fixed across regimes, *differences* between curves identify differences in the learned response functions, while *levels* inherit the reference population’s covariate mix and should be read only relative to one another. SATO-conditioned partial dependence (replication materials) confirms the composition mechanism of [Landy \(2024\)](#): within every regime, higher-SATO tertiles trace steeper curves. The decomposition thus reconciles the composition view with the observation of [Katz and Li \(2024\)](#) that late-2025 realized S-curves exceed pandemic-era steepness: both are features of the same estimated function.

Attribution analysis (mean absolute SHAP values; replication materials) tells the same story at the level of drivers: pre-pandemic prepayment is dominated by burnout and seasoning; the wave by the incentive; and in the lock-in era all drivers compress, consistent with the decline in achievable discrimination.

5.4 Scenario read-through for the current coupon stack

Finally, we apply the full-sample model to the December 2025 cross-section of active 2023–24 vintage 6.0–7.0% coupons under parallel shifts of the mortgage rate, holding burnout and house prices fixed (a deliberately transparent one-month-hazard exercise, annualized). Table 3 reports projected speeds.

Cohort	Unchanged	–50bp	–100bp	–150bp
2023 6.0s	6.9	12.9	20.7	25.9
2023 6.5s	13.1	22.5	29.1	36.0
2023 7.0s	25.4	34.4	43.7	52.0
2024 6.0s	6.5	12.1	19.4	26.1
2024 6.5s	13.3	23.0	32.4	42.1
2024 7.0s	25.5	35.5	43.7	52.2

Table 3: Model-projected CPR for the December 2025 cross-section of 2023–24 vintage cohorts under parallel mortgage-rate shifts.

The 7.0% coupons are already fast at unchanged rates (≈ 25 CPR) and reach refinancing-wave speeds (above 50 CPR) under a 150 basis point rally; the 6.5s approximately triple over the same interval and constitute the most rate-sensitive segment of the stack. This is the model-implied counterpart of the record negative convexity measured in the current-coupon market in late 2025 ([Katz and Li, 2024](#)), localized to specific vintage-coupon cohorts.

6 Limitations

Our sample is Fannie Mae collateral only; Ginnie Mae collateral, whose VA segment exhibited the fastest and most policy-sensitive speeds of the 2024–25 episodes, is outside the dataset. The pre-pandemic regime is observed for only two years (vintages 2018+). The scenario analysis is a one-month hazard annualized with burnout and house prices held fixed; it understates path dependence in a sustained rally and is not a valuation (OAS) model. The level of the 2020–21 wave is under-predicted by all specifications, ours and commercial alike, reflecting absent capacity and media-effect covariates. Partial-dependence curves are counterfactuals on a fixed population, not cohort forecasts. Finally, monthly PMMS timing and the publication lag of quarterly house-price indices introduce weeks-level measurement slack, disclosed here and in the replication materials.

7 Conclusion

Using open data and commodity hardware, we provide a public model-side account of the 2022 mortgage prepayment regime break. The prepayment function did not merely shift: its refinancing limb steepened while its turnover component became governed by the prevailing-rate level. A model frozen at the break never recovered (its cohort errors remained above three CPR points in every post-break year), while annual re-estimation restored sub-point accuracy only after roughly two years of new-regime data. For practitioners, the results argue for explicit regime-awareness in model governance: recalibration works, but its speed limit is data accumulation, which favors supplementing re-estimation with regime-conditioning or judgmental overlays in the interim. They also argue for close attention to the steep embedded optionality of the 2023–24 high-coupon stack. For researchers, the removal-record leakage taxonomy of Section 3.4 is a cautionary catalogue for anyone estimating event models on the public agency datasets: an out-of-sample AUC of 0.9999 is not a discovery.

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A Robustness

This appendix reports four checks on the empirical machinery. Unless stated otherwise, they use the gradient-boosted model estimated through December 2024 and evaluated on 2025, the split with the largest training window.

A.1 Calibration

Figure 3 plots, by decile of predicted hazard, the mean predicted monthly hazard against the design-weighted empirical event rate on the held-out year. The middle deciles lie close to the diagonal; the model modestly under-predicts in the slower deciles (e.g., 0.13% predicted versus 0.15% realized in the first decile) and over-predicts in the fastest decile (2.86% versus 2.51%). Fitted hazards are thus usable as levels, not merely as rankings, with mild shrinkage toward the center at both tails.

A.2 Cohort-level fit

Figure 4 plots predicted against actual CPR for the 754 vintage $\times$ coupon cohort-months (with beginning balance above \$1bn) of the 2025 test year; marker area is proportional to cohort balance. The UPB-weighted mean absolute error is 0.86 CPR points. Predictions track actuals closely through roughly 20 CPR, where nearly all of the balance resides; the fastest cohort-months are modestly attenuated (predicted below actual above approximately 25 CPR), the cohort-level counterpart of the tail shrinkage in Figure 3, so point forecasts for the fastest, smallest cohorts should be read as conservative. (The 0.86 figure equals the 0.9 entry of Table 1 up to the \$1bn cohort filter and rounding.)

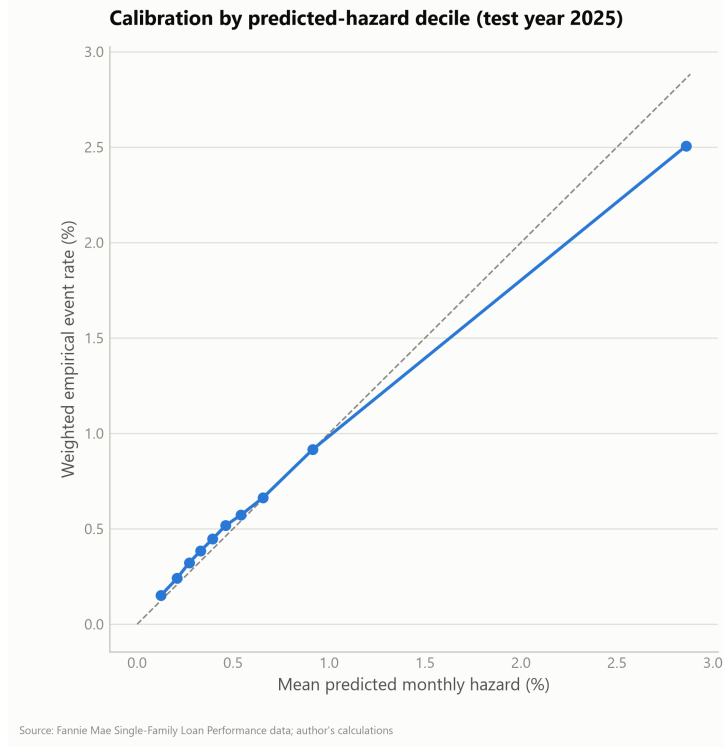


Figure 3: Reliability diagram on the 2025 test year: mean predicted monthly hazard versus design-weighted empirical event rate, by predicted-hazard decile. Dashed line is the 45-degree diagonal.

A.3 Sensitivity to the non-event sampling rate

The estimation panel retains non-event loan-months with probability $\kappa = 5\%$ (Section 4.1). Re-estimating the December 2024 model after deterministically halving the retained non-events (an effective κ of 2.5%, weights doubled accordingly) leaves results essentially unchanged (Table 4), consistent with the Horvitz–Thompson argument: the downsampling rate affects efficiency, not identification.

Panel	Training rows	AUC (2025)	Cohort CPR MAE (pts)
Baseline, $\kappa = 5\%$	2,037,093	0.691	0.86
Thinned, $\kappa = 2.5\%$	1,206,248	0.690	0.88

Table 4: Sensitivity of the December 2024 model to the non-event retention rate. Evaluation on all 2025 months.

A.4 S-curve robustness across SATO buckets

Figure 5 repeats the partial-dependence analysis of Section 5.3 within tertiles of spread at origination. The regime ordering of Section 5.3 is preserved within every tertile, and within every regime higher-SATO tertiles trace steeper curves, the composition mechanism emphasized by Landy (2024), confirming that the S-curve rotation is not an artifact of pooling across SATO.

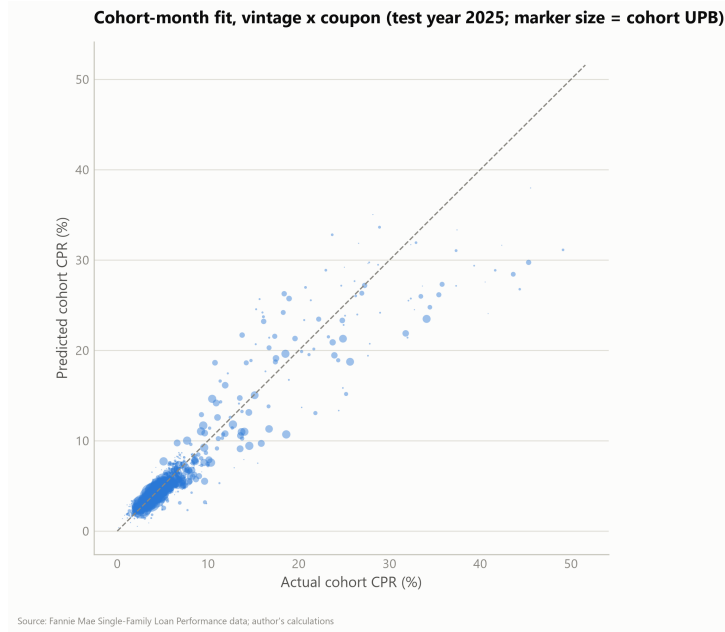


Figure 4: Predicted versus actual cohort CPR, 2025 test year, vintage $\times$ coupon cohort-months with beginning UPB above \$1bn. Marker area proportional to cohort balance.

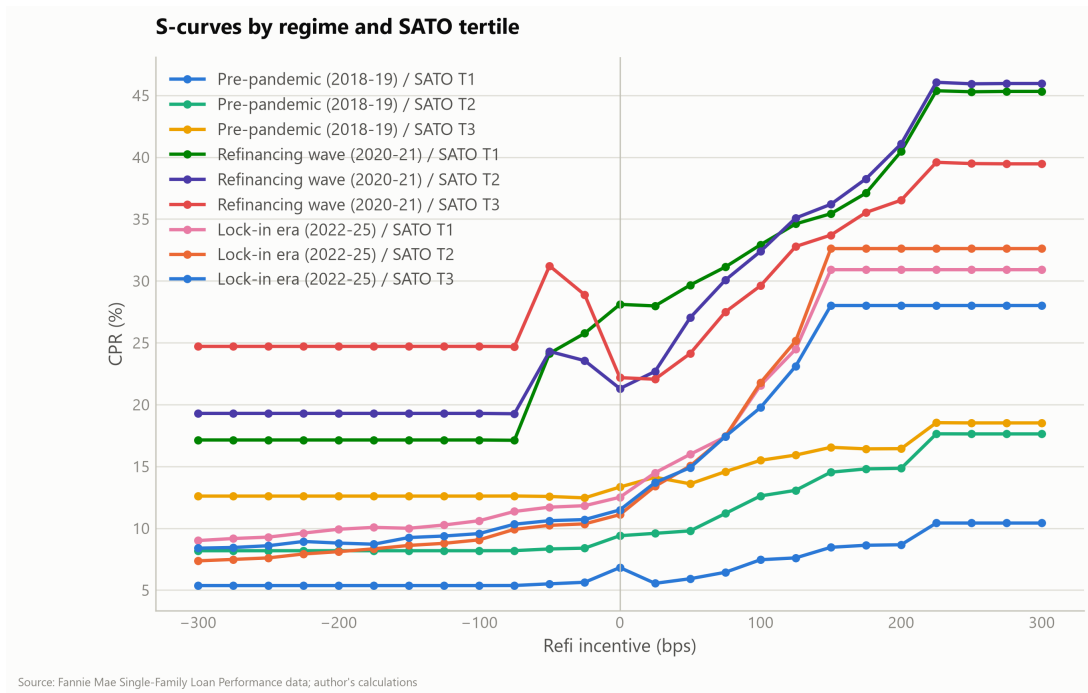


Figure 5: Model-implied S-curves by regime and SATO tertile on the common reference population.